

Rarity, Compression, and Meaning in Finite Program Spaces

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Exact atlases of Boolean formulas and sorting networks

Abstract

We test an operational creativity triad—correctness, compression, and low prior probability—in two finite spaces that permit exact enumeration. All 256 three-input Boolean functions are synthesized in NAND, NOR, and AND/OR/NOT formula languages. Syntax-free semantic descriptors initially predict minimum formula size with leave-one-function-out $R^2 = 0.831\text{--}0.920$. The result survives stricter validation: after holding out all functions with the same invariant descriptor vector, $R^2 = 0.613\text{--}0.656$, beyond 1,000 null permutations matched on bias, algebraic degree, and symmetry ($p < 0.001$ in all languages). Meaning therefore constrains synthesis difficulty. It does not determine a representation-free score: NAND–NOR rank correlations are 0.548 for minimum size, 0.376 for boundary surprisal, and 0.264 for canonical compression gain. Exhaustive enumeration of 221,832 correct four-input sorting networks gives a complementary resolution test. The 12 shortest networks share one entropy curve but follow 12 distinct full output-distribution trajectories. At lengths six through eight, entropy distinguishes 27, 147, and 497 trajectories, whereas the full representation distinguishes 756, 4,356, and 15,036. Semantic structure is predictive, but apparent rigidity, rarity, and compression depend on the language, prior, and observational resolution.

Question and operational formalism. The motivating proposal assigns a candidate program e a conjunctive score

$$\mathcal{K}(e) = \underbrace{\mathcal{S}(e)}_{\text{structure or function}} \underbrace{\mathcal{G}(e)}_{\text{preserved}} \underbrace{\mathcal{R}(e)}_{\text{compression}} \underbrace{\mathcal{P}(e)}_{\text{gain low prior probability}}. \quad (1)$$

The product notation is logical rather than psychometric: no factor is meant to substitute for another. The finite experiments make each dependency explicit. For a language $\mathcal{L}$, let $E_{\ell}^{\mathcal{L}}$ be the ordered formulas or programs of exact length ℓ , and let $n_{\ell}^{\mathcal{L}}(f)$ count those computing semantic function f . Define

$$L_{\mathcal{L}}^*(f) = \min\{\ell : n_{\ell}^{\mathcal{L}}(f) > 0\}, \quad R_{\mathcal{L}}(f) = -\log_2 \frac{n_{L_{\mathcal{L}}^*(f)}^{\mathcal{L}}(f)}{|E_{L_{\mathcal{L}}^*(f)}^{\mathcal{L}}|}, \quad (2)$$

and $G_{\mathcal{L}}(f) = L_{0,\mathcal{L}}(f) - L_{\mathcal{L}}^*(f)$ relative to a frozen canonical compiler L_0 . Correctness is exhaustive over the complete finite input domain. Minimality and rarity are exact under the declared grammar and the uniform prior conditional on length.

Boolean universe and semantic map. The Boolean universe contains every function $f : \{0, 1\}^3 \rightarrow \{0, 1\}$. Semantic dynamic programming counts ordered formulas by exact truth-table semantics through 14 gates; all 256 functions are reached in each language. Counts for single-binary-gate NAND and NOR are checked against $\text{Catalan}(\ell)3^{\ell+1}$, and the mixed grammar is checked against its independent unary/binary recurrence.

The syntax-free descriptor $\psi(f)$ contains signed output bias, sorted variable influences, Walsh–Fourier energy at degrees zero through three, algebraic degree, and the size of the input-permutation stabilizer. It is invariant to renaming the three inputs. There are 36 exact descriptor vectors; their two-dimensional principal-component projection occupies 22 visible positions. Figure 1 holds the coordinates fixed and changes only the overlaid exact minimum size.

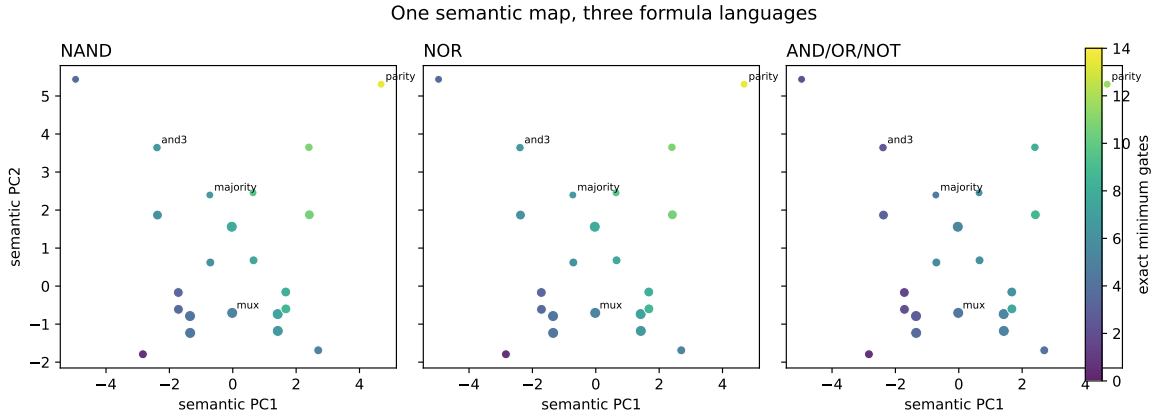


Figure 1: A common semantic map colored by exact minimum formula size in three languages. Point area records projected multiplicity. Parity is difficult in all three languages, but the color changes elsewhere show that semantic location does not uniquely determine syntactic cost. The map is descriptive; all tests use the full invariant descriptor vector.

Representation stability. If complexity were intrinsic to the truth-table function, rankings would be stable under reasonable complete bases. Table 1 shows a mixed result. Minimum-size ranks are strongly, though imperfectly, aligned between NAND or NOR and the mixed basis. NAND and NOR agree much less when the same function is held fixed. Boundary surprisal is less stable, and canonical compression gain is least stable. The polarity of the primitive operation and the compiler baseline materially change the numerical claim.

Language pair	Minimum size	Boundary surprisal	Compression gain
NAND / NOR	0.548	0.376	0.264
NAND / AND-OR-NOT	0.805	0.739	0.612
NOR / AND-OR-NOT	0.805	0.739	0.601

Table 1: Spearman rank correlations across all 256 functions. Persistence is evidence for a semantic component; disagreement quantifies dependence on the chosen representation and prior.

Stress test 1: group-held-out prediction. The optimistic leave-one-function-out estimate can leak semantic twins: an input-permuted copy of the test function may remain in training. We therefore use seven-nearest-neighbor prediction under three increasingly strict protocols. The first removes only the target function; the second removes its entire orbit under S_3 input permutations (80 orbits); the third removes its entire exact ψ -descriptor class (36 classes). Features are standardized once without using synthesis targets. Predictions are pooled over the 256 held-out functions.

Language	Function held out	Orbit held out	Descriptor held out	Strict RMSE
NAND	0.831	0.726	0.613	1.583
NOR	0.833	0.729	0.643	1.522
AND-OR-NOT	0.920	0.859	0.656	1.161

Table 2: Held-out R^2 for minimum gate count; “Strict RMSE” corresponds to descriptor-class holdout. Eliminating semantic twins lowers the estimate substantially but leaves a majority of variance predictable.

Stress test 2: matched nulls. High prediction could merely restate elementary facts such as output balance, algebraic degree, or symmetry. Within each exact stratum of signed bias, degree, and stabilizer size, we permute minimum formula sizes and rerun the strict descriptor-class protocol 1,000 times. The null mean and central 95% interval are -0.064 $[-0.271, 0.118]$ for NAND, -0.068 $[-0.269, 0.106]$ for NOR, and 0.014 $[-0.149, 0.155]$ for AND/OR/NOT. No null replicate reaches its observed R^2 , giving the finite-simulation bound $p = 1/1001 < 0.001$ in each language. Figure 2 displays both the attenuation under stricter holdouts and the null separation.

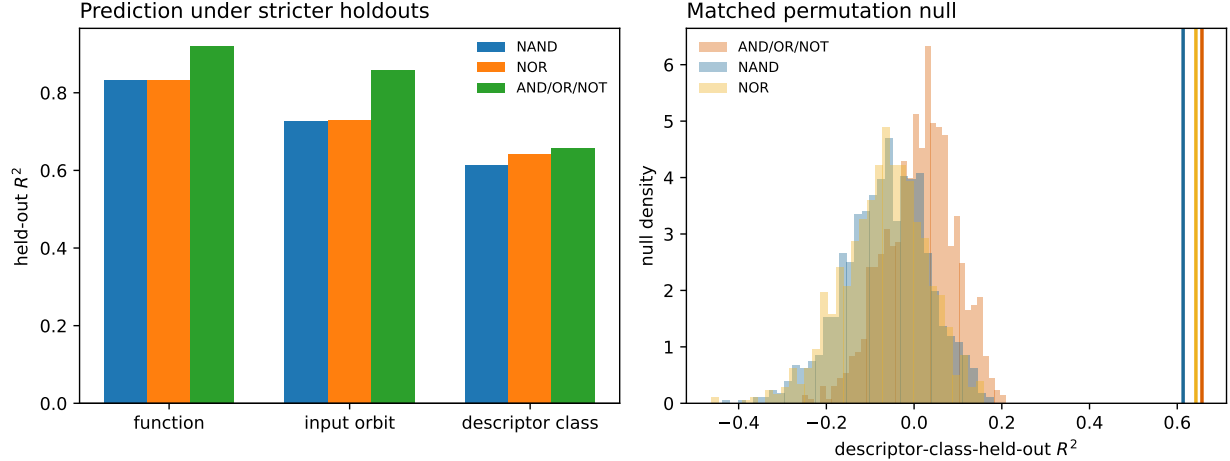


Figure 2: Boolean validation. Left: prediction weakens when whole semantic groups are excluded but remains positive. Right: vertical lines are the observed strict R^2 values; histograms are null distributions preserving bias, degree, and symmetry. The surviving signal must come from finer influence/Fourier structure or correlated semantic organization.

Sorting universe. A four-input comparator network is a word over the six unordered wire pairs. All 24 input permutations are evaluated. No network of length at most four sorts; exactly 12 of $6^5 = 7,776$ length-five words sort, giving 9.340 bits of boundary surprisal. The correct counts at lengths six, seven, and eight are 900, 16,800, and 204,120. Correctness therefore becomes more common away from the compression boundary. Moreover, 8.0%, 32.1%, and 49.4% of correct networks at those lengths reach the sorted state early and append redundant comparators.

For prefix $e_{1:t}$, let Y_t be its output on a uniformly sampled input permutation and let

$$p_t(y) = \Pr(Y_t = y), \quad H_t = - \sum_y p_t(y) \log_2 p_t(y). \quad (3)$$

The first atlas represented a trajectory only by $(H_0, \dots, H_L)$. This coarse statistic yielded one profile for all 12 optimal networks, suggesting a rigid boundary.

Stress test 3: full permutation-state geometry. Entropy forgets which output permutations remain. The refined representation retains the complete 24-coordinate vector p_t at every prefix. A single global wire permutation $\pi \in S_4$ acts on all p_t ; a trajectory is canonicalized by the lexicographically least member of this 24-element orbit:

$$[P(e)] = \min_{\pi \in S_4} (\pi p_0, \pi p_1, \dots, \pi p_L). \quad (4)$$

The quotient produces no additional mergers among correct paths because the fixed ascending target anchors the wire order. Table 3 gives the resulting resolution comparison. The original rigidity claim does not survive: the 12 optima have the same entropy curve but 12 distinct full paths. Nevertheless the boundary remains small in absolute terms, and full-path diversity grows rapidly with redundant length.

Length	Correct words	Entropy paths	Full paths / wires	Raw full paths
5	12	1	12	12
6	900	27	756	756
7	16,800	147	4,356	4,356
8	204,120	497	15,036	15,036

Table 3: Exact sorting-network trajectory counts. Several comparator words can induce the same semantic path. Entropy collapses many paths further and therefore overstates semantic rigidity.

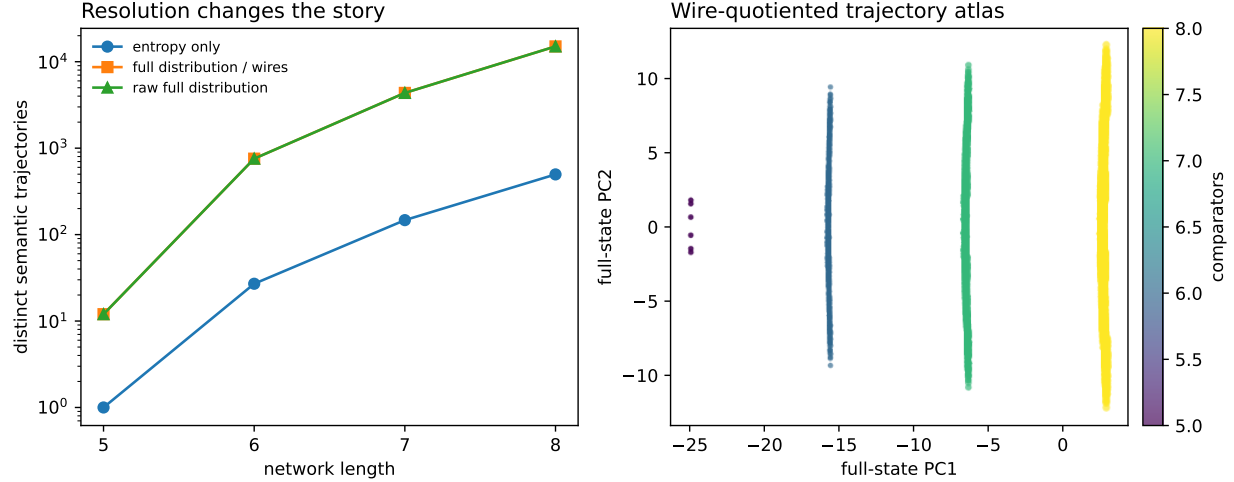


Figure 3: Sorting validation. Left: trajectory diversity under coarse entropy, full output distributions modulo global wire relabeling, and raw full distributions. Right: a descriptive PCA of resampled canonical full trajectories; color records program length and point area records multiplicity. The separated length bands partly reflect normalization to a common time grid, so inference uses exact counts rather than the projection.

What the experiments establish. Three conclusions survive the stress tests. First, semantic structure has substantial predictive content for exact synthesis difficulty beyond trivial symmetry classes. Second, the values assigned to rarity and compression are not properties of a function alone; they are conditional on a language, compiler baseline, and prior. Third, a compressed boundary can be both rare and constrained while still containing distinctions hidden by a coarse semantic statistic. “One trajectory” was a statement about entropy, not about computation in full permutation space.

These conclusions support a qualified version of (1). The triad is useful only with subscripts: $\mathcal{G}_{\mathcal{L}, L_0}$ and $\mathcal{R}_{\mathcal{L}, P}$. A credible semantic claim should additionally be stable across a declared family of representations and observational scales. The experiments do not measure human originality, discovery effort, aesthetic value, or a universal Kolmogorov complexity. They verify finite operational proxies and show exactly where those proxies fail to be intrinsic.

Implication for formal proofs. For Lean, the result argues against interpreting isolation in one embedding as mathematical profundity. A stronger test would represent both theorem statement and proof journey at several semantic resolutions, remove entire equivalence neighborhoods rather than individual declarations, and require a signal to survive vocabulary ablation, repository matching, and alternative formal encodings. The controlled universes demonstrate that such a semantic signal can exist; they also demonstrate that its numerical strength will be overstated unless representation dependence and resolution are tested directly.

Reproducibility. All enumeration, invariants, group assignments, null draws (seed 0), tables, and figures are generated by the checked-in scripts. The Boolean synthesis covers all 256 functions; sorting counts are independently checked against the exact totals above. The validation requires no cloud services or stochastic model API. The archived source essay and machine-readable CSV/JSON artifacts are retained alongside the experiments.